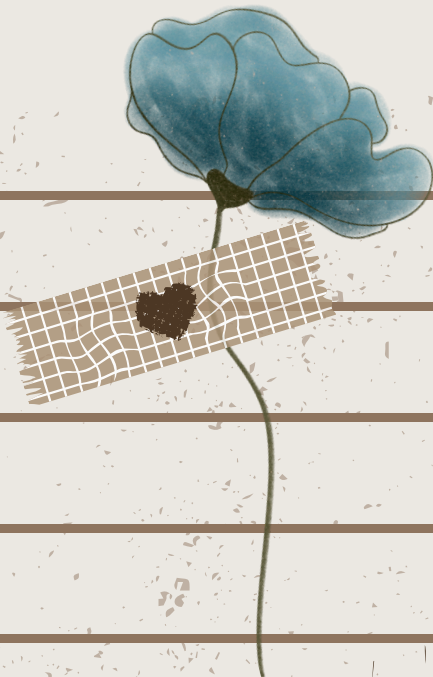




DiGiTAL **SiGNA** **P R o C E S S I N G** **F i N A L s**

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Z-TRANSFORM

- The z-transform of a discrete time signal $x(n)$ is defined as the power series

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

- Where z is a complex variable. It is sometimes called the direct z-transform because it transforms the time domain signal $x(n)$ into its complex-plane representation $X(z)$.
- The ROC (region of convergence) of $X(z)$ is the set of all values of z for which $X(z)$ attains a finite value.

Z-transform



$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$z = re^{j\theta}$ - polar form of complex variable z

r - magnitude of the complex variable z

$r = |z| = \text{mod } z$

$\theta = \text{complex argument or phase or angle}$

Examples



1 $x[n] = \left(\frac{1}{2}\right)^n u[n]$

$$\begin{aligned} x[z] &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n (1)z^{-n} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \end{aligned}$$

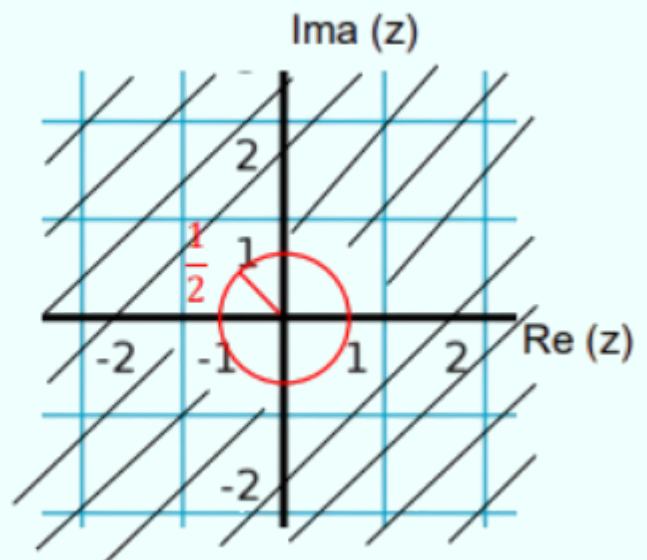
$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

ROC:

$$\left|\frac{1}{2}z^{-1}\right| < 1$$

$$\left|\frac{1}{2z}\right| < 1$$

$$|z| > \frac{1}{2}$$



$$x[z] = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

2

$$x[n] = u[n]$$

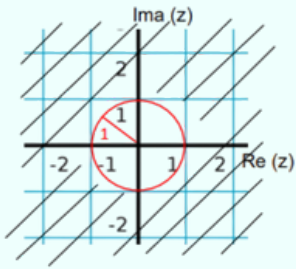
$$= \sum_{n=-\infty}^{\infty} (1)z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^n,$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$x[z] = \frac{1}{1-z^{-1}}$$

ROC:
 $|z^{-1}| < 1$
 $\left|\frac{1}{z}\right| < 1$
 $|z| > 1$



3

$$x[n] = -u[-n-1]$$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} (-1)z^{-n}$$

$$= - \sum_{n=1}^{\infty} (z)^n, \quad n = -n$$

$$= - \sum_{m=0}^{\infty} (z)^{m+1}, \quad \begin{matrix} n-1 = m \\ n = m+1 \end{matrix}$$

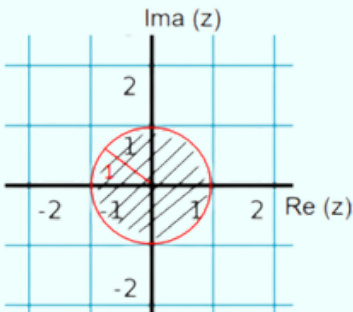
$$= - \sum_{m=0}^{\infty} z^m z$$

$$= -z \sum_{m=0}^{\infty} z^m, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$= \frac{-z}{1-z} = \left[\frac{-z}{1-z}\right] \frac{-z}{-z}$$

$$x[z] = \frac{1}{1-\frac{1}{z}}$$

ROC:
 $|z^{-1}| < 1$
 $\left|\frac{1}{z}\right| < 1$
 $|z| < 1$



4

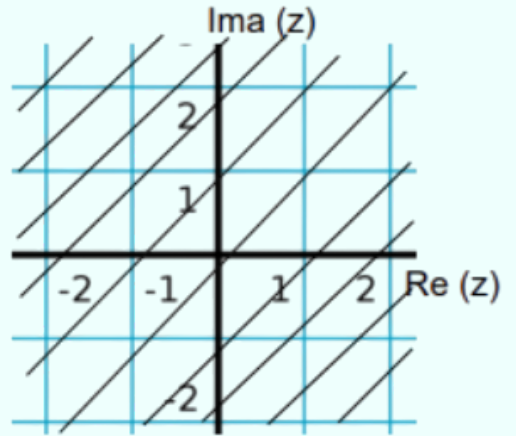
$$x[n] = \delta[n]$$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

$$= \sum_{n=-\infty}^{-1} \delta[n]z^{-n}$$

$$= z^{-n}|_{n=0}$$

$$x[z] = z^0 = 1$$



5

$$x[n] = \delta[n-k]$$

$$x[z] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

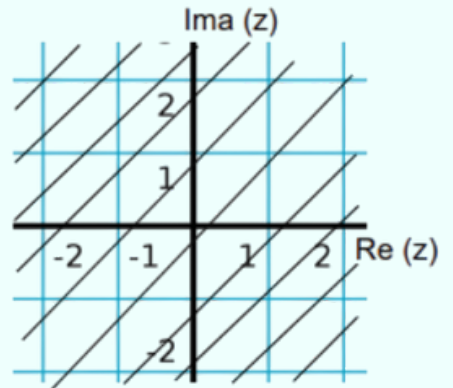
$$= \sum_{n=-\infty}^{-1} \delta[n]z^{-n}$$

$$= z^{-n}|_{n=k}$$

$$x[z] = z^{-k}$$

$z = 0$ when $k > 0$

$z = \infty$ when $k < 0$

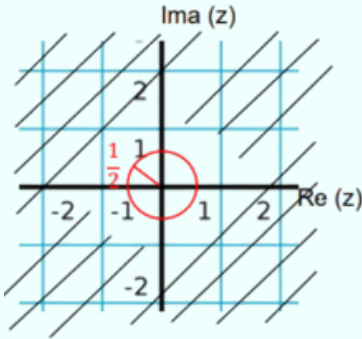


6

$$\begin{aligned}
 x[n] &= 7\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n u[n] \\
 x[z] &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\
 &= \sum_{n=0}^{\infty} \left[7\left(\frac{1}{3}\right)^n - 6\left(\frac{1}{2}\right)^n\right] z^{-n} \\
 &= 7 \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n} - 6 \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} \\
 &= 7 \sum_{n=0}^{\infty} \left(\frac{1}{3}z^{-1}\right)^n - 6 \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}
 \end{aligned}$$

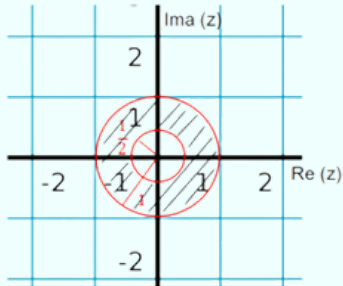
$$x[z] = \frac{7}{1 - \frac{1}{3}z^{-1}} - \frac{6}{1 - \frac{1}{2}z^{-1}}$$

$$\begin{aligned}
 \text{ROC}_1: \quad & \left|\frac{1}{3}z^{-1}\right| < 1 & \text{ROC}_2: \quad & \left|\frac{1}{2}z^{-1}\right| < 1 \\
 & \left|\frac{1}{3z}\right| < 1 & & \left|\frac{1}{2z}\right| < 1 \\
 & |z| > \frac{1}{3} & & |z| > \frac{1}{2}
 \end{aligned}$$



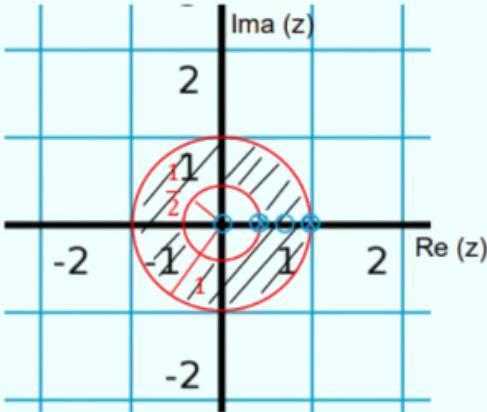
$$\begin{aligned}
 \text{ROC}_1: \quad & |z^{-1}| < 1 & \text{ROC}_2: \quad & \left|\frac{1}{2}z^{-1}\right| < 1 \\
 & \left|\frac{1}{z}\right| < 1 & & \left|\frac{1}{2z}\right| < 1 \\
 & |z| < 1 & & |z| > \frac{1}{2}
 \end{aligned}$$

$$\text{ROC: } \frac{1}{2} < |z| < 1$$



$$\begin{aligned}
 x[z] &= \left[\frac{1}{1 - \frac{1}{z}} + \frac{1}{1 - \frac{1}{2}z^{-1}} \right] \frac{z}{z} \\
 &= \frac{z}{z-1} + \frac{z}{z-\frac{1}{2}} \\
 &= \frac{z\left(z-\frac{1}{2}\right) + z(z-1)}{(z-1)\left(z-\frac{1}{2}\right)} \\
 &= \frac{z^2 - \frac{1}{2}z + z^2 - z}{(z-1)\left(z-\frac{1}{2}\right)} \\
 &= \frac{2z^2 - \frac{3}{2}z}{(z-1)\left(z-\frac{1}{2}\right)} \\
 &= \frac{z\left(2z - \frac{3}{2}\right)}{(z-1)\left(z-\frac{1}{2}\right)}
 \end{aligned}$$

Zeroes	Poles
$z = 0$	$z - 1 = 0$
$2z - \frac{3}{2} = 0$	$z = \frac{1}{2}$
$z = \frac{3}{4}$	$z - \frac{1}{2} = 0$
	$z = \frac{1}{2}$



7

$$\begin{aligned}
 x[n] &= -u[-n-1] + \left(\frac{1}{2}\right)^n u[n] \\
 x[z] &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\
 &= \sum_{n=-\infty}^{\infty} \left[-u[-n-1] + \left(\frac{1}{2}\right)^n u[n]\right] z^{-n} \\
 &= - \sum_{n=-\infty}^{-1} z^{-n} + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}, \quad n = -n \\
 &= - \sum_{m=0}^{\infty} z^m + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \quad \begin{matrix} n-1=m \\ n=m+1 \end{matrix} \\
 &= - \sum_{m=0}^{\infty} z^{m+1} + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n \\
 &= - \sum_{m=0}^{\infty} z^m z + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n \\
 &= -z \sum_{m=0}^{\infty} z^m + \sum_{n=0}^{\infty} \left(\frac{1}{2}z^{-1}\right)^n, \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \\
 &= \frac{-z}{1-z} + \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{-z}{-z}\left[\frac{-z}{1-z}\right] + \frac{1}{1-\frac{1}{2}z^{-1}} \\
 &= \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}}
 \end{aligned}$$

$$x[z] = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}}$$

8

$$x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n]$$

$$= \sum_{n=0}^{\infty} \left[\left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n] \right] z^{-n}, \quad \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n \left(\frac{e^{\frac{j\pi n}{4}} - e^{-\frac{j\pi n}{4}}}{2j} \right) z^{-n}$$

$$= \frac{1}{2j} \sum_{n=0}^{\infty} \left(\frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}\right)^n - \frac{1}{2j} \sum_{n=0}^{\infty} \left(\frac{1}{3} e^{-\frac{j\pi}{4}} z^{-1}\right)^n$$

$$x[z] = \frac{1}{2j} \left(\frac{1}{1 - \frac{1}{3} e^{\frac{j\pi}{4}} z^{-1}} \right) - \frac{1}{2j} \left(\frac{1}{1 - \frac{1}{3} e^{-\frac{j\pi}{4}} z^{-1}} \right)$$

ROC₁:

$$\left| \frac{1}{3} e^{\frac{j\pi}{4}} z^{-1} \right| < 1$$

$$e^{\pm\theta} = \cos \theta \pm j \sin \theta$$

$$e^{\frac{j\pi}{4}} = \cos \frac{\pi}{4} + j \sin \frac{\pi}{4}$$

$$= \frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2}$$

$$\left| e^{\frac{j\pi}{4}} \right| = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = 1$$

$$\left| \frac{1}{3z} \right| < 1$$

$$|z| > \frac{1}{3}$$

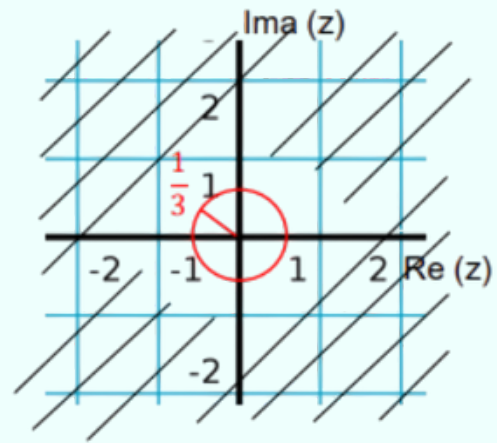
ROC₂:

$$\left| \frac{1}{3} e^{-\frac{j\pi}{4}} z^{-1} \right| < 1$$

$$e^{\pm\theta} = \cos \theta \pm j \sin \theta$$

$$\left| \frac{1}{3z} \right| < 1$$

$$|z| > \frac{1}{3}$$



INVERSE Z-TRANSFORM



- The inverse Z-transform is defined as the process of finding the time domain signal $x[n]$ from its Z-transform $X(z)$.

When using the **z-transform**

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

it is often useful to be able to find $x[n]$ given $X(z)$. There are at least 4 different methods to do this:

1. Inspection
2. Partial-Fraction Expansion
3. Power Series Expansion
4. Contour Integration



1

$$X(z) = \frac{\frac{1}{4}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{4}z^{-1})}$$

$$X(z) = \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - \frac{1}{4}z^{-1}}$$

$$A = \frac{\frac{1}{4}z^{-1}}{1 - \frac{1}{4}z^{-1}} \Big|_{z^{-1}=2} = \frac{\frac{1}{4}(2)}{1 - \frac{1}{4}(2)} = \frac{0.5}{1 - 0.5} = \frac{0.5}{0.5} = 1$$

$$B = \frac{\frac{1}{4}z^{-1}}{1 - \frac{1}{2}z^{-1}} \Big|_{z^{-1}=4} = \frac{\frac{1}{4}(4)}{1 - \frac{1}{2}(4)} = \frac{1}{1 - 2} = \frac{1}{-1} = -1$$

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - \frac{1}{4}z^{-1}}$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{4}\right)^n u[n]$$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] - \left[-\left(\frac{1}{4}\right)^n u[-n-1]\right]$$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] + \left(\frac{1}{4}\right)^n u[-n-1]$$

$$x[n] = \underbrace{-\left(\frac{1}{2}\right)^n u[-n-1]}_{\text{Pole } 1/2 \text{ (Anti-causal)}} - \underbrace{\left(\frac{1}{4}\right)^n u[n]}_{\text{Pole } 1/4 \text{ (Causal)}}$$

2

$$X(z) = \frac{3 + 2z^{-1} + z^{-2}}{1 - 3z^{-1} + 2z^{-2}}$$

$$1 - 3z^{-1} + 2z^{-2} = (1 - z^{-1})(1 - 2z^{-1})$$

$$X(z) = C + \frac{A}{1 - z^{-1}} + \frac{B}{1 - 2z^{-1}}$$

$$C = \frac{\text{Coeff of } z^{-2} \text{ in Num}}{\text{Coeff of } z^{-2} \text{ in Den}} = \frac{1}{2} = 0.5$$

$$A = \frac{3 + 2z^{-1} + z^{-2}}{1 - 2z^{-1}} \Big|_{z^{-1}=1} = \frac{3 + 2(1) + 1}{1 - 2(1)} = \frac{6}{-1} = -6$$

$$B = \frac{3 + 2z^{-1} + z^{-2}}{1 - z^{-1}} \Big|_{z^{-1}=0.5} = \frac{3 + 2(0.5) + (0.5)^2}{1 - 0.5} = \frac{3 + 1 + 0.25}{0.5} = \frac{4.25}{0.5} = 8.5$$

$$X(z) = 0.5 - \frac{6}{1 - z^{-1}} + \frac{8.5}{1 - 2z^{-1}}$$

$$x[n] = 0.5\delta[n] - 6(1)^n u[n] + 8.5(2)^n u[n]$$

3

$$X(z) = \frac{1 + 3z^{-1}}{1 + 3z^{-1} + 2z^{-2}}$$

$$X(z) = \frac{(1 + 3z^{-1}) \cdot z^2}{(1 + 3z^{-1} + 2z^{-2}) \cdot z^2}$$

$$X(z) = \frac{z^2 + 3z}{z^2 + 3z + 2}$$

$$X(z) = \frac{z(z + 3)}{z^2 + 3z + 2}$$

$$\frac{X(z)}{z} = \frac{z + 3}{z^2 + 3z + 2}$$

$$z^2 + 3z + 2 = (z + 1)(z + 2)$$

$$\frac{X(z)}{z} = \frac{z + 3}{(z + 1)(z + 2)} = \frac{A}{z + 1} + \frac{B}{z + 2}$$

$$A = \left. \frac{z + 3}{z + 2} \right|_{z=-1} = \frac{-1 + 3}{-1 + 2} = \frac{2}{1} = 2$$

$$B = \left. \frac{z + 3}{z + 1} \right|_{z=-2} = \frac{-2 + 3}{-2 + 1} = \frac{1}{-1} = -1$$

$$\frac{X(z)}{z} = \frac{2}{z + 1} + \frac{-1}{z + 2}$$

$$x[n] = 2(-1)^n u[n] - (-2)^n u[n]$$

5

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$\text{Remainder} = (1 + 2z^{-1} + z^{-2}) - 2 \left(1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2} \right)$$

$$\text{Remainder} = (1 + 2z^{-1} + z^{-2}) - (2 - 3z^{-1} + z^{-2})$$

$$\text{Remainder} = (1 - 2) + (2 - (-3))z^{-1} + (1 - 1)z^{-2}$$

$$\text{Remainder} = -1 + 5z^{-1}$$

$$X(z) = 2 + \frac{-1 + 5z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$8u[n] - 9 \left(\frac{1}{2} \right)^n u[n]$$

$$x[n] = 2\delta[n] + 8u[n] - 9 \left(\frac{1}{2} \right)^n u[n]$$

4

$$X(z) = \frac{-1 + 5z^{-1}}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$X(z) = \frac{(-1 + 5z^{-1}) \cdot z^2}{(1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}) \cdot z^2}$$

$$X(z) = \frac{-z^2 + 5z}{z^2 - \frac{3}{2}z + \frac{1}{2}}$$

$$X(z) = \frac{z(-z + 5)}{z^2 - \frac{3}{2}z + \frac{1}{2}}$$

$$z^2 - \frac{3}{2}z + \frac{1}{2} = (z - 1)(z - \frac{1}{2})$$

$$\frac{X(z)}{z} = \frac{-z + 5}{(z - 1)(z - \frac{1}{2})} = \frac{A}{z - 1} + \frac{B}{z - \frac{1}{2}}$$

$$A = \left. \frac{-z + 5}{z - \frac{1}{2}} \right|_{z=1} = \frac{-1 + 5}{1 - 0.5} = \frac{4}{0.5} = 8$$

$$B = \left. \frac{-z + 5}{z - 1} \right|_{z=0.5} = \frac{-0.5 + 5}{0.5 - 1} = \frac{4.5}{-0.5} = -9$$

$$\frac{X(z)}{z} = \frac{8}{z - 1} + \frac{-9}{z - \frac{1}{2}}$$

$$x[n] = 8u[n] - 9 \left(\frac{1}{2} \right)^n u[n]$$

INVERSE Z-TRANSFORM USING POWER SERIES FUNCTION

- If ROC is greater than any number it is called casual. The denominator must be arranged from highest power to lowest power.
- If ROC is less than any number it is called non-casual. The denominator must be arranged from lowest to highest power.

Power Series?



Example



1 Find the IZT of $X(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}}$ ROC: $|z| > 1$

$$\begin{array}{r} 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} \\ 1 - 1.5z^{-1} + 0.5z^{-2} \overline{) 1} \\ \underline{1 - 1.5z^{-1} + 0.5z^{-2}} \\ 1.5z^{-1} - 0.5z^{-2} \\ \underline{1.5z^{-1} - 2.25z^{-2} + 0.75z^{-3}} \\ 1.75z^{-2} - 0.75z^{-3} \\ \underline{1.75z^{-2} - 2.625z^{-3} + 0.875z^{-4}} \\ 1.875z^{-3} - 0.875z^{-4} \end{array}$$

$$x[z] = 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} + \dots$$

$$x[n] = \delta[n] + 1.5\delta[n-1] + 1.75\delta[n-2] + 1.875\delta[n-3] + \dots$$

$$x[n] = \{1, 1.5, 1.75, 1.875, \dots\}$$

2 Find the IZT of $X(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}}$ ROC: $|z| < 0.5$

$$\begin{array}{r} 2z^2 + 6z^3 + 14z^4 + 30z^5 \\ 0.5z^{-2} - 1.5z^{-1} + 1 \overline{) 1} \\ \underline{1 - 3z + 2z^2} \\ 3z - 2z^2 \\ \underline{3z - 9z^2 + 6z^3} \\ 7z^2 - 6z^3 \\ \underline{7z^2 - 21z^3 + 14z^4} \\ 15z^3 - 14z^4 \end{array}$$

$$x[z] = 2z^2 + 6z^3 + 14z^4 + 30z^5 + \dots$$

$$x[n] = 2\delta[n+2] + 6\delta[n+3] + 14\delta[n+4] + 30\delta[n+5] + \dots$$

$$x[n] = \{\dots, 30, 14, 6, 2, 0, 0\}$$

3 Find the IZT of $X(z) = \frac{z}{2z^2-3z+1}$ ROC: $|z| > \frac{1}{2}$

$$\begin{array}{r} \frac{1}{2}z^{-1} + \frac{3}{4}z^{-2} + \frac{7}{8}z^{-3} \\ 2z^2 - 3z + 1 \overline{) z} \\ \underline{z - \frac{3}{2} + \frac{1}{2}z^{-1}} \\ \frac{3}{2} - \frac{1}{2}z^{-1} \\ \underline{\frac{3}{2} - \frac{9}{4}z^{-1} + \frac{3}{4}z^{-2}} \\ \frac{7}{4}z^{-1} - \frac{3}{4}z^{-2} \\ \underline{\frac{7}{4}z^{-1} - \frac{21}{8}z^{-2} + \frac{7}{8}z^{-3}} \\ \frac{15}{8}z^{-2} - \frac{7}{8}z^{-3} \end{array}$$

$$x[z] = \frac{1}{2}z^{-1} + \frac{3}{4}z^{-2} + \frac{7}{8}z^{-3} + \dots$$

$$x[n] = \frac{1}{2}\delta[n-1] + \frac{3}{4}\delta[n-2] + \frac{7}{8}\delta[n-3] + \dots$$

$$x[n] = \left\{0, \frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \dots\right\}$$

4 $X(z) = \frac{z}{2z^2-3z+1}$ ROC: $|z| > \frac{1}{2}$

$$\begin{array}{r} z + 3z^2 + 7z^3 \\ 1 - 3z + 2z^2 \overline{) z} \\ \underline{z - 3z^2 + 2z^3} \\ 3z^2 - 2z^3 \\ \underline{3z^2 - 9z^3 + 6z^4} \\ 7z^3 - 6z^4 \\ \underline{7z^3 - 21z^4 + 14z^5} \\ 15z^4 - 14z^5 \end{array}$$

$$x[z] = z + 3z^2 + 7z^3 + \dots$$

$$x[n] = \delta[n+1] + 3\delta[n+2] + 7\delta[n+3] + \dots$$

$$x[n] = \{\dots, 7, 3, 1, 0\}$$

FILTERS

Filters!



- series of components designed to reduce the amplitude or attenuate of an unwanted frequency in respect to some cutoff frequency

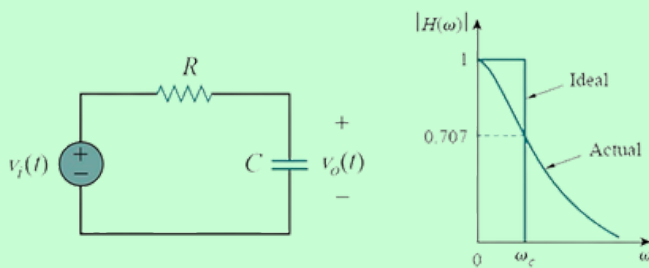
- passes lower frequencies and stops high frequencies

Low Pass



Lowpass Filters

A **lowpass filter** is designed to pass only frequencies from dc up to the cutoff frequency ω_c



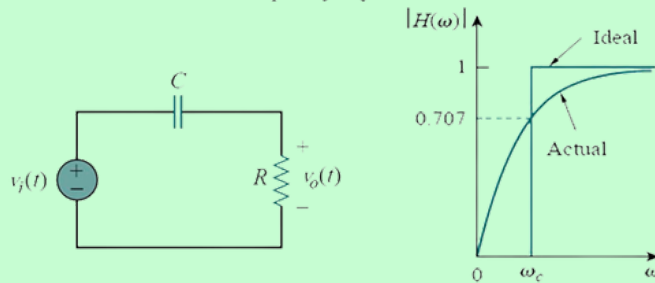
High Pass



- pass high frequencies and rejects low frequencies

Highpass Filter

A **highpass filter** is designed to pass all frequencies above its cutoff frequency ω_c



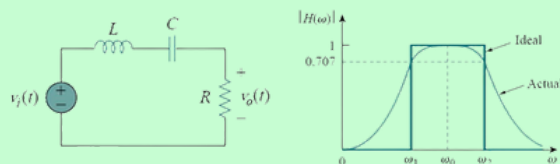
Bandpass



- passes frequencies within a frequency band and blocks or attenuates frequencies outside the band

Bandpass Filter

A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega_0 < \omega_2$



Bandstop



- passes frequencies outside a frequency band and blocks or attenuates frequencies within the band

Bandstop Filter

A **bandstop filter** is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega_0 < \omega_2$

